

Technical and Executive Report: Model Accuracy Assessment

1. Executive Summary

This report provides an initial technical assessment of the predictive accuracy of the machine learning model `main.py` (v1) using the supplied training dataset. The model applies a Random Forest Regression approach to estimate a continuous target variable ("value") using categorical and date-derived features.

The analysis process included:

- Loading and preprocessing the training dataset
- Converting the target field into a numeric format
- Engineering date-related variables ("year" and "month")
- Applying one-hot encoding to categorical variables
- Splitting the dataset into training and testing subsets
- Training a Random Forest Regressor
- Evaluating performance using Mean Absolute Error (MAE)

The model uses the following predictor variables:

- description
- geography
- sex
- age
- source_file
- year
- month

A train-test split of 80% training data and 20% testing data was applied using a fixed random seed (`random_state=42`) to support reproducibility.

The primary limitation of this assessment is the absence of model output metrics and supporting validation information. As a result, no conclusions can currently be drawn regarding whether model performance is acceptable for operational or policy use.

2. Background and Purpose

The purpose of this analysis was to assess the predictive accuracy of the model `main.py` (v1) using the supplied training data.

The modelling workflow was implemented in Python using:

- pandas
- scikit-learn

The model selected for this analysis was a `RandomForestRegressor`, a supervised ensemble learning method commonly used for regression problems involving structured datasets.

Assumption:

It is assumed that the target variable "value" represents a continuous numerical measure suitable for regression modelling.

3. Methodology Summary

Data Preparation:

- Conversion of the target variable (“value”) into numeric format
- Conversion of the “date” column into datetime format
- Engineering of temporal variables (“year” and “month”)
- One-hot encoding of categorical variables

Model Development:

- Train-test split: 80% training / 20% testing
- RandomForestRegressor with 200 estimators
- random_state=42

Evaluation:

- Mean Absolute Error (MAE)

[INFORMATION NEEDED: Final MAE value and any additional evaluation metrics]

4. Key Findings

- The modelling approach follows a standard supervised machine learning workflow.
- Date feature engineering was applied to improve baseline predictive capability.
- Invalid target and date rows were removed before training.
- The model evaluation used Mean Absolute Error (MAE).

However:

- No MAE result was provided.
- No benchmark or comparison model was supplied.
- No cross-validation evidence was included.

[INFORMATION NEEDED: MAE result, comparison baseline, validation results, and interpretation criteria]

5. Limitations and Caveats

The following limitations apply:

- The dataset was described as “trimmed”, which may reduce representativeness and introduce selection bias.
- Only a single evaluation metric (MAE) was used.
- The methodology used a single train-test split rather than cross-validation.
- Important contextual information such as dataset size and performance thresholds was not supplied.
- One-hot encoding may create high-dimensional sparse feature sets depending on category volume.

6. Appendix

Appendix A — Model Specification

Model name: main.py (v1)

Model type: Random Forest Regressor

Estimators: 200

Train-test split: 80/20

Random seed: 42

Evaluation metric: Mean Absolute Error (MAE)

Appendix B — Predictor Variables

- description
- geography
- sex
- age
- source_file
- year
- month